

THESIS SEMINAR

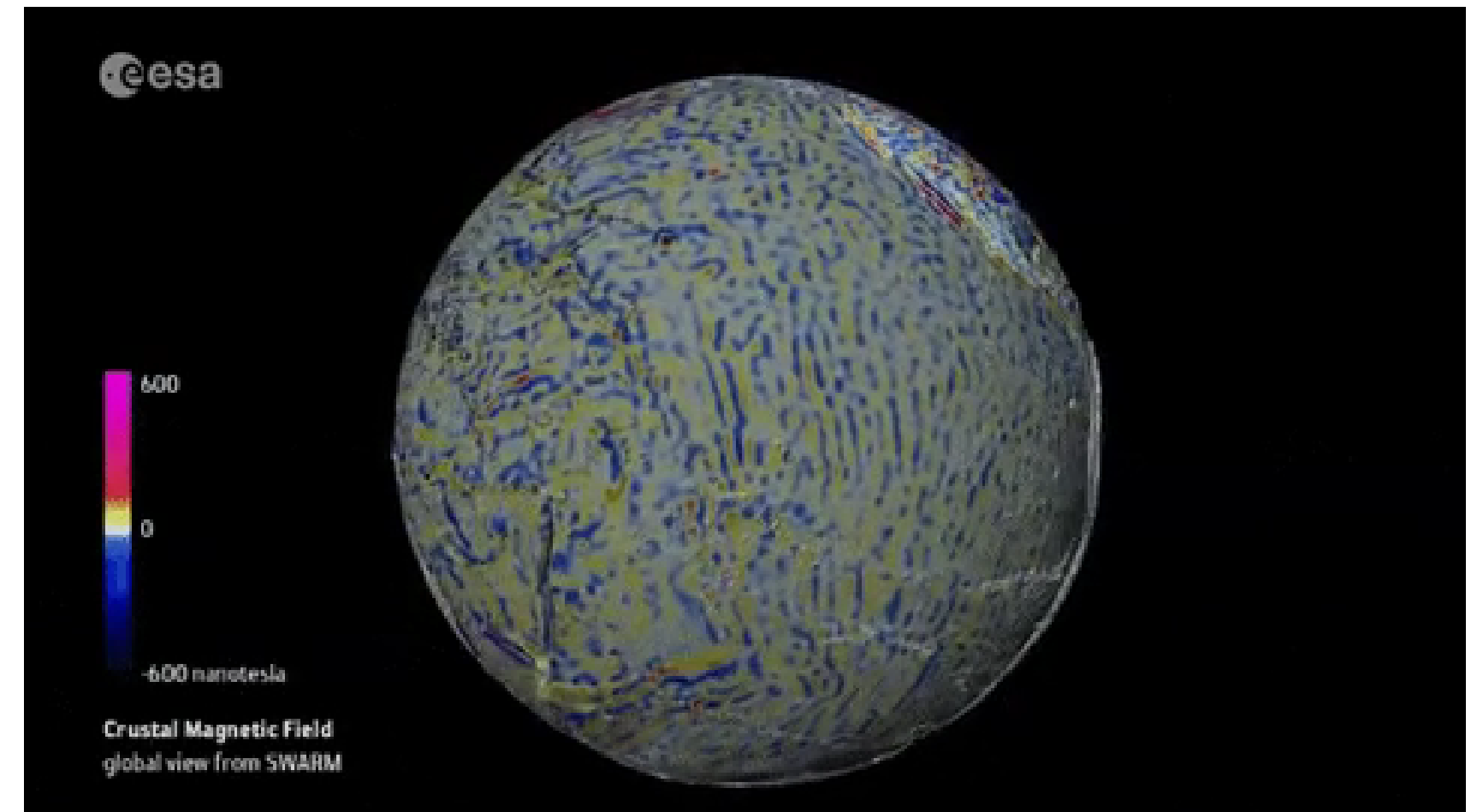
15-10-2025

Generation of optical RS data based on
radar RS data using generative AI.

(SAR-to-Optical Image Translation)

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OVERVIEW

- Background
- Problem
- Solutions
- Methodology
- Results
- Ablation Studies
- Challenges
- Limitations & Future work
- Summary

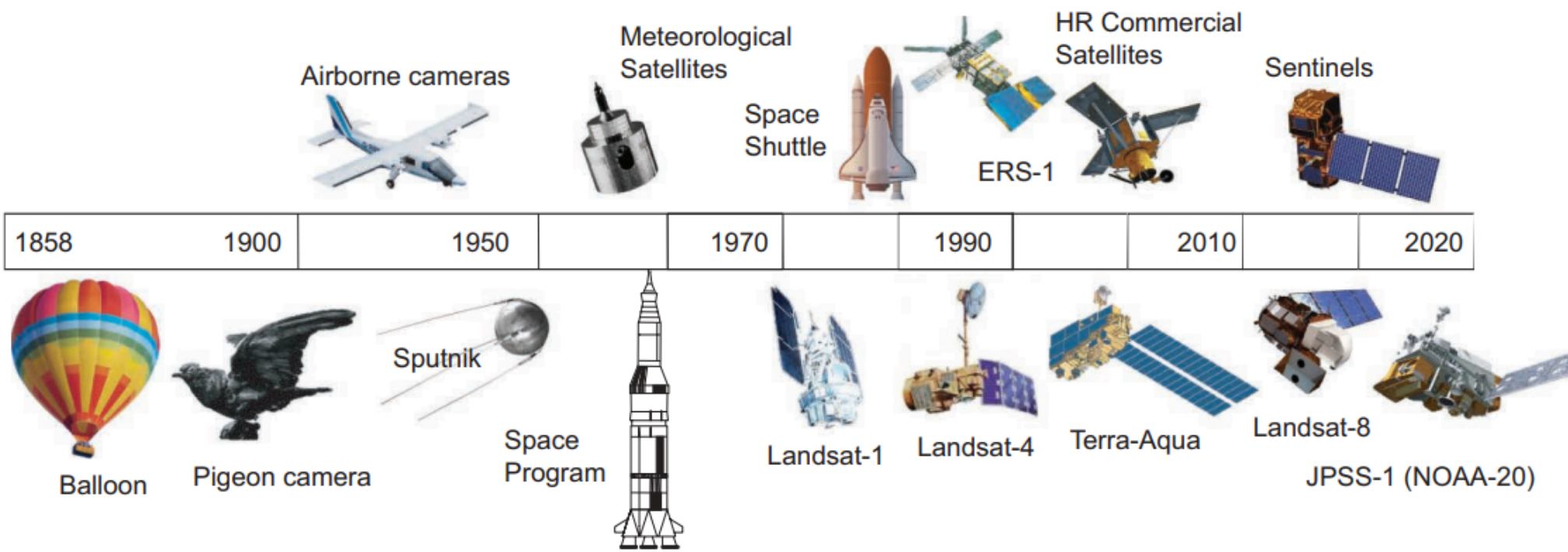


Remote Sensing

- The acquisition of information about an object through sensors without direct physical contact.
- Interdisciplinary nature: electromagnetic – spectroscopy – physics – geology
- Term introduced in early 1960s
- Origin of RS: invention of photography in 1839

RS Applications

- Atmospheric dynamics
- Cloud distribution
- Vegetation cover (seasonal variability)
- Polar ice dynamics
- Tropical deforestation
- Wildfire detection



Fact

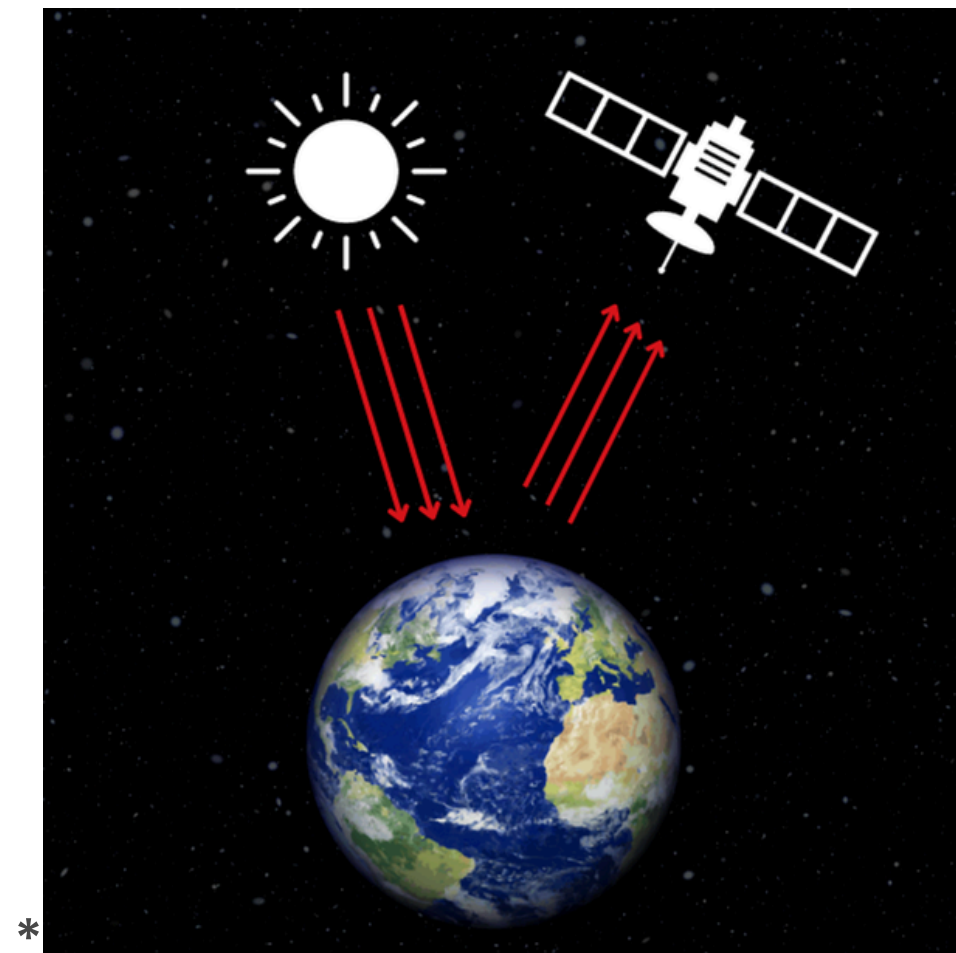
To date, every planet in the solar system has been visited at least once



Sensors

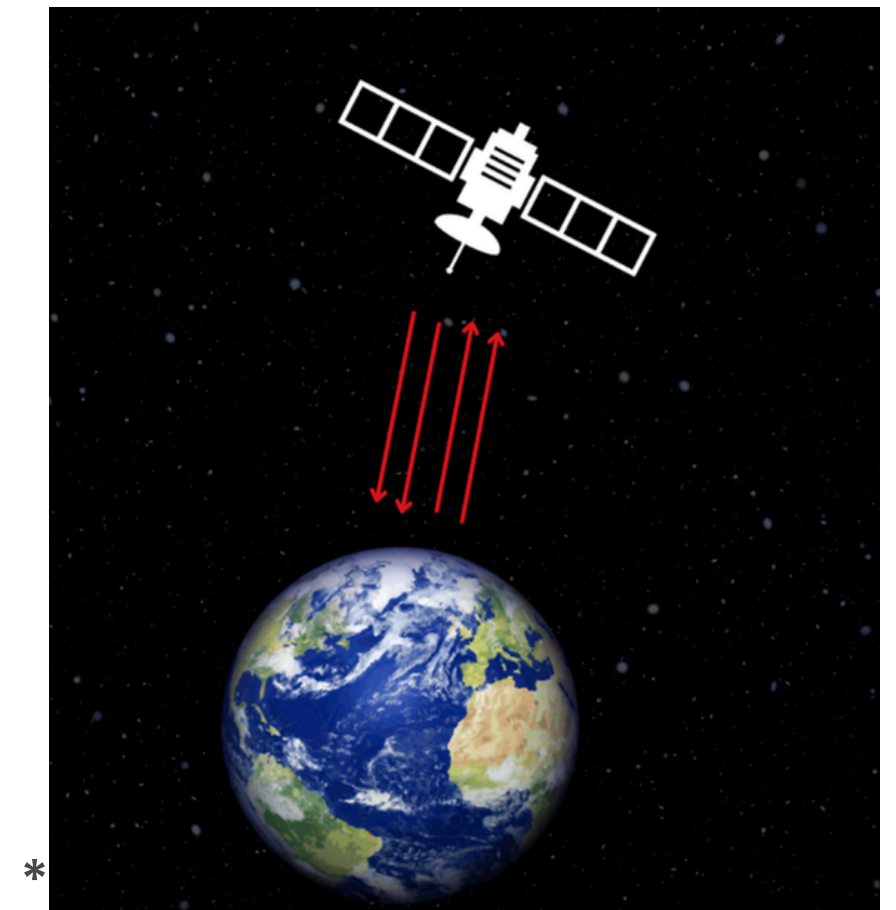
Passive

- External or natural source, e.g. solar radiation
- Ex: optical sensors – Landsat



Active

- Emit own energy pulses to illuminate the target and measure reflected signals.
- Ex: Radar, LiDAR

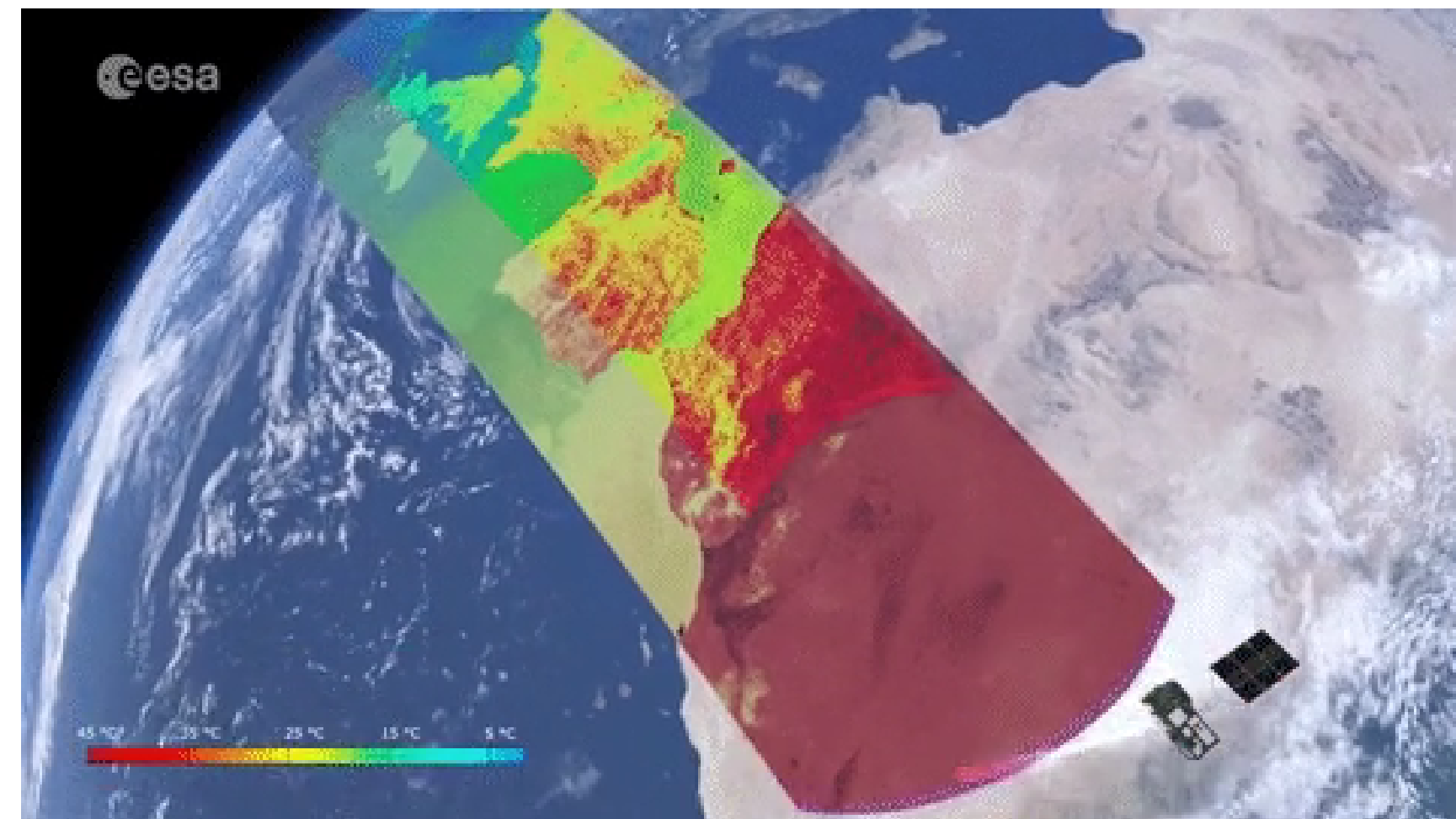


Copernicus: Europe's eyes on Earth

- Earth observation component of the European Union's Space Programme
- Funded, coordinated, and managed by the European Commission in co-opwith partners, e.g. ESA

Services:

- land
- marine
- atmosphere
- emergency management
- security
- climate change



Copernicus: Europe's eyes on Earth

SENTINEL-3

- 2016 & 2018
- Ocean and Land

SENTINEL-4

- 2025
- UV, visible, NIR



SENTINEL-5P

- 2017
- Atmosphere

SENTINEL-6

- 2020
- Sea surface topography

Copernicus: Europe's eyes on Earth

SENTINEL-1

- Two sun-synchronous, near-polar orbiting satellites in the same orbital plane, separated by 180° in phase
- Sentinel-1A (2014) – Sentinel-1C (2024)
- C-band synthetic aperture radar (SAR) operating at 5.4 GHz
- All-weather, day-and-night imaging
- Single (HH or VV) and dual (HH+HV or VV+VH) polarisation
- Global coverage within 6 days

APPLICATIONS

- Detection of land subsidence
- Structural deformation
- Ground movements



Copernicus: Europe’s eyes on Earth

SENTINEL-2

- Two sun-synchronous, near-polar orbiting satellites in the same orbital plane, separated by 180° in phase
- Sentinel-2A (2015) – Sentinel-2B (2017) – Sentinel-2C (2024)
- High-resolution optical imagery over land and coastal regions
- global coverage within 5 days at the Equator

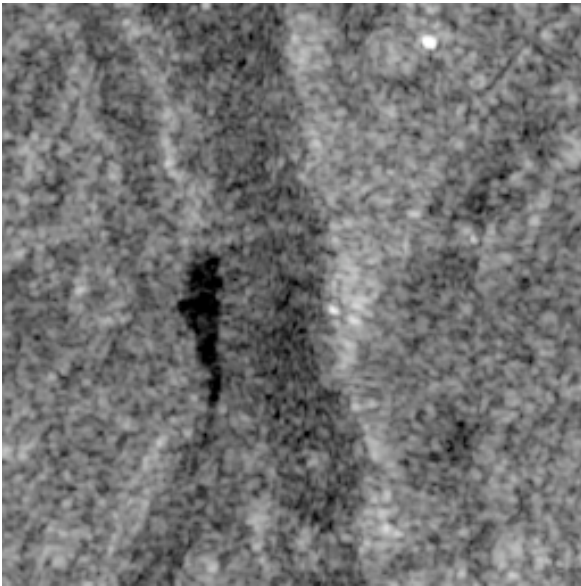


APPLICATIONS

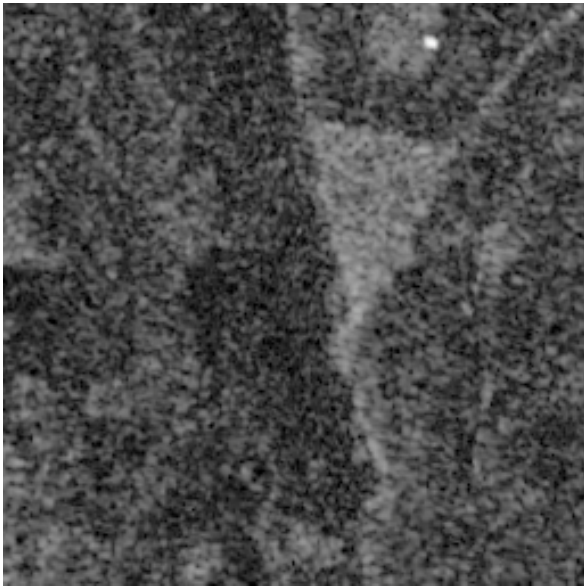
- Vegetation and crop monitoring
- Land cover mapping
- Water quality assessment
- Snow and ice monitoring
- Cloud screening
- Atmospheric correction

Band	Central Wavelength [nm]	Resolution [m]
B1	443 (Aerosols)	60
B2	490 (Blue)	10
B3	560 (Green)	10
B4	665 (Red)	10
B5	705 (Red edge)	20
B6	740 (Red edge)	20
B7	783 (Red edge)	20
B8	842 (NIR)	10
B8a	865 (Red edge)	20
B9	945 (Water vapour)	60
B10	1375 (Cirrus)	60
B11	1610 (SWIR)	20
B12	2190 (SWIR)	20

SAR data (Sentinel-1)

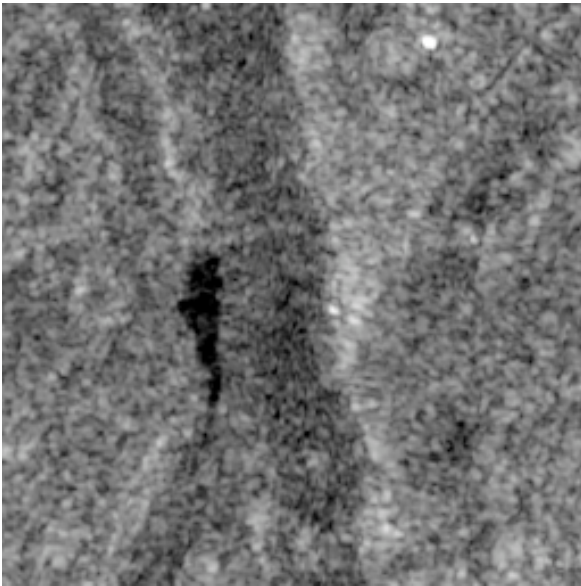


VV

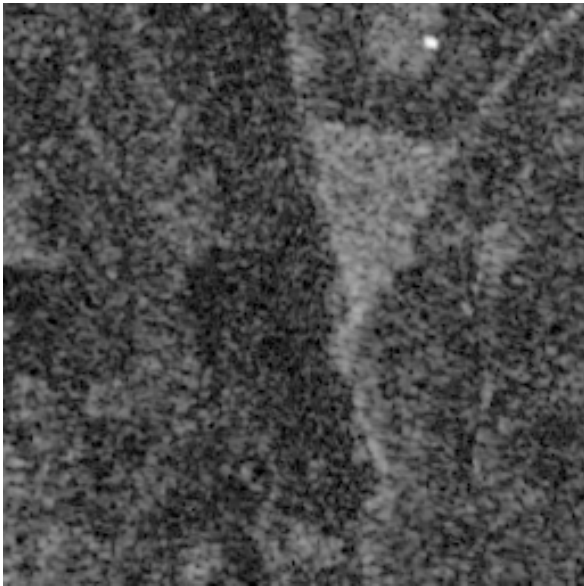


VH

SAR data (Sentinel-1)



VV



VH

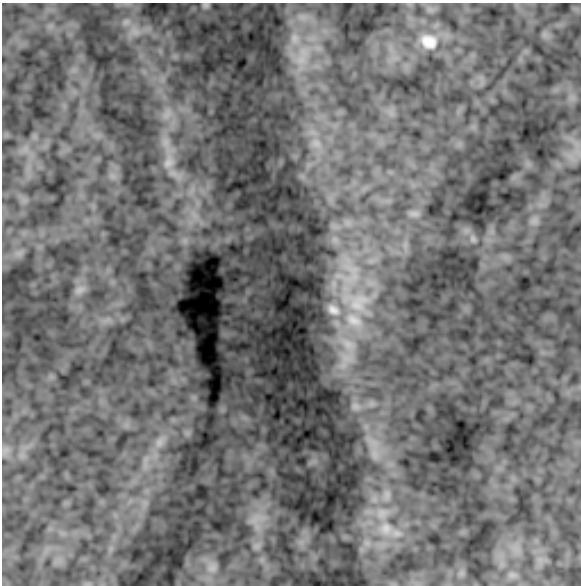
Optical data (Sentinel-2)



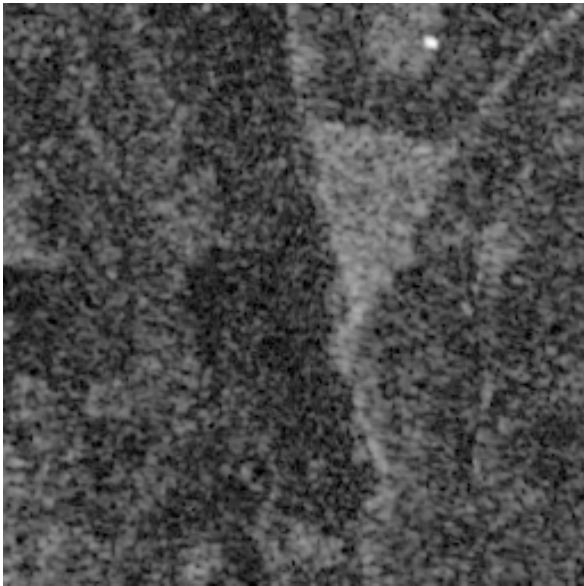
cloud-free

Cloud Cover

SAR data (Sentinel-1)



VV

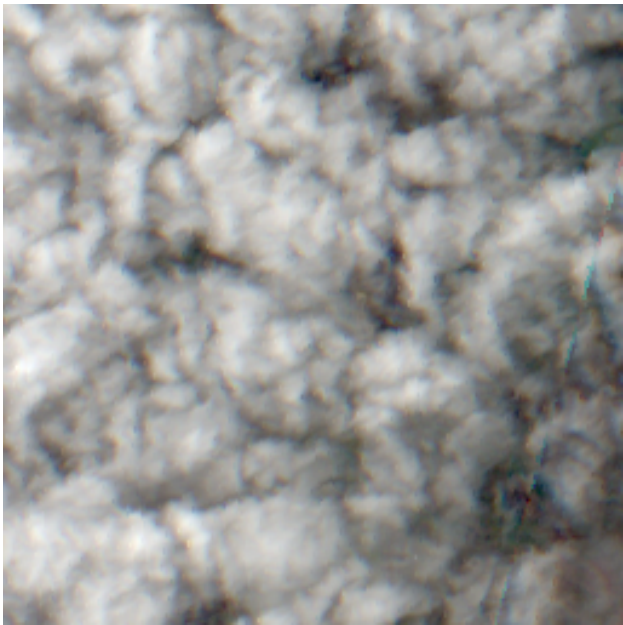


VH

Optical data (Sentinel-2)



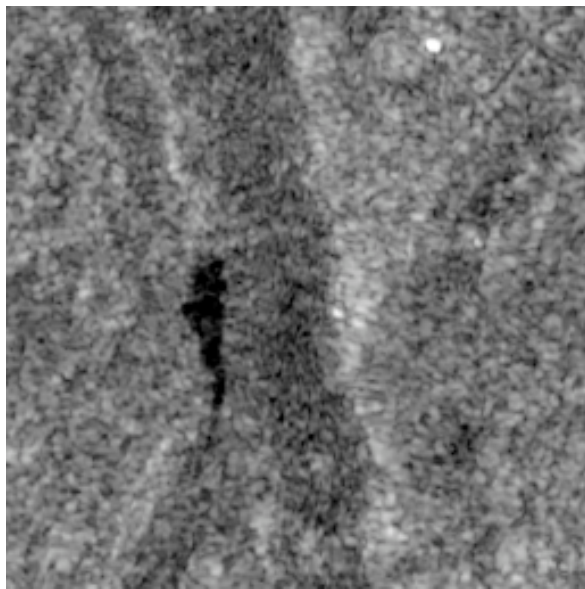
cloud-free



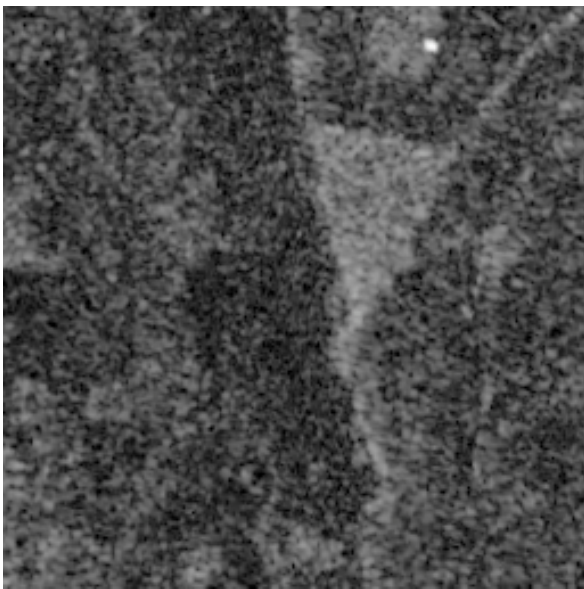
cloudy

Cloud Cover

SAR data (Sentinel-1)



VV



VH

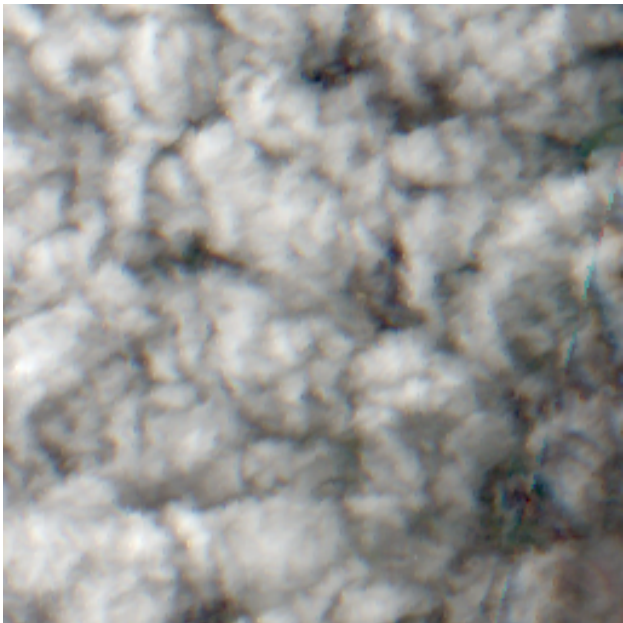
✓ All-weather, day & night acquisition

✗ Interpretability

Optical data (Sentinel-2)



cloud-free



cloudy

✓ Interpretability

✗ Cloud cover
global cloud cover surpasses 66%, 55% over land

Cloud Removal

1. Single-image methods

- Statistical & physical approaches: spatial similarity, frequency filtering, atmospheric scattering models
- CNNs, U-Nets, and GAN-based model



Thin or semi-transparent clouds



Dense cloud cover

2. Multimodal-based methods:

- Integrate auxiliary data from other sensors to improve optical image restoration.
- Ex: Multispectral-based methods exploit the differential sensitivity of spectral bands
- Ex: Fusing synthetic aperture radar (SAR) with optical imagery (**SAR-to-optical translation**)



Dense cloud removal



Data registration



Modality difference

3. Multitemporal-based methods:

- Repeated acquisitions of the same location at different times to fill in cloud-covered areas



Dense cloud removal



Geometric misalignment



Temporal variability in land cover

Cloud Removal

Category	Advantages	Limitations	Representative literature
Single-image	• No auxiliary data required (cost- and time-efficient). • Effective for thin or semi-transparent clouds. • Straightforward implementation with statistical/physical models or deep learning.	• Ineffective for dense or opaque clouds. • Often introduces artifacts or color distortions. • Deep learning requires large paired datasets, which are difficult to obtain.	[11] [12] [20] [21] [22] [23] [24] [25] [26]
Multimodal	• Integrates complementary information from other sensors. • Multispectral bands provide spectral redundancy. • SAR-optical fusion enables SAR-to-optical translation, penetrating cloud layers. • Suitable for both thin and thick clouds.	• Requires accurate registration of heterogeneous data. • SAR data introduces speckle noise. • High computational complexity and preprocessing effort.	[13] [27] [28] [9] [29] [30] [31] [32] [33] [34] [35] [36]
Multitemporal	• Exploits temporal redundancy to reconstruct cloudy regions. • Effective for dense and extensive cloud cover. • Deep learning models can capture spatio-temporal correlations. • Can be extended with SAR-optical time series for improved robustness.	• Sensitive to geometric misalignment and temporal variability. • Requires consistent multitemporal datasets, which may be unavailable. • Landscape or seasonal changes reduce restoration accuracy.	[28] [37] [15] [16] [38] [39]

Research Gaps

Majority of studies translated only 3 RGB bands (B2, B3, B4), some with NIR (B8). Few (4 works) do the full bands

- L. Abady et al. [7] → Poor validation, quantitative evaluation, no across band, no SAR involved, no cloud removal validation
- MTcGAN [4] → only 9 bands, only 1000 samples, missing quantitative evaluation, spatio-temporal limitation (subarea of Illinois within the US Corn Belt, summer 2022), no across band evaluation
- S2MS-GAN [5] → different data, no across band evaluation, poor evaluation
- DSen2-CR [6] → solid methodology, trained on SEN12-MS-CR, semi-good results

- 1- Validate the SAR-to-optical image translation for the whole spectrum (13 bands)**
- 2- How reliably each optical band can be individually reconstructed and to which extent**
- 3- The performance of SAR-to-optical image translation in cloud removal**

Problem Formulation

Frame SAR-to-optical translation as a supervised image-to-image regression task using a conditional GAN framework.

Input: $X \in \mathbb{R}^{H \times W \times 2}$

Output: $Y \in \mathbb{R}^{H \times W \times N}$

Objective: Learn a mapping from Sentinel-1 SAR images to Sentinel-2 multispectral optical images.

$$f_{\theta} : \mathbb{R}^{H \times W \times 2} \rightarrow \mathbb{R}^{H \times W \times N}$$

so that the generated multispectral image $\hat{Y} = f_{\theta}(X)$ is as close as possible to the reference image Y

Training this model requires minimizing a loss function $\mathcal{L}$ that measures the discrepancy between $\hat{Y}$ and Y :

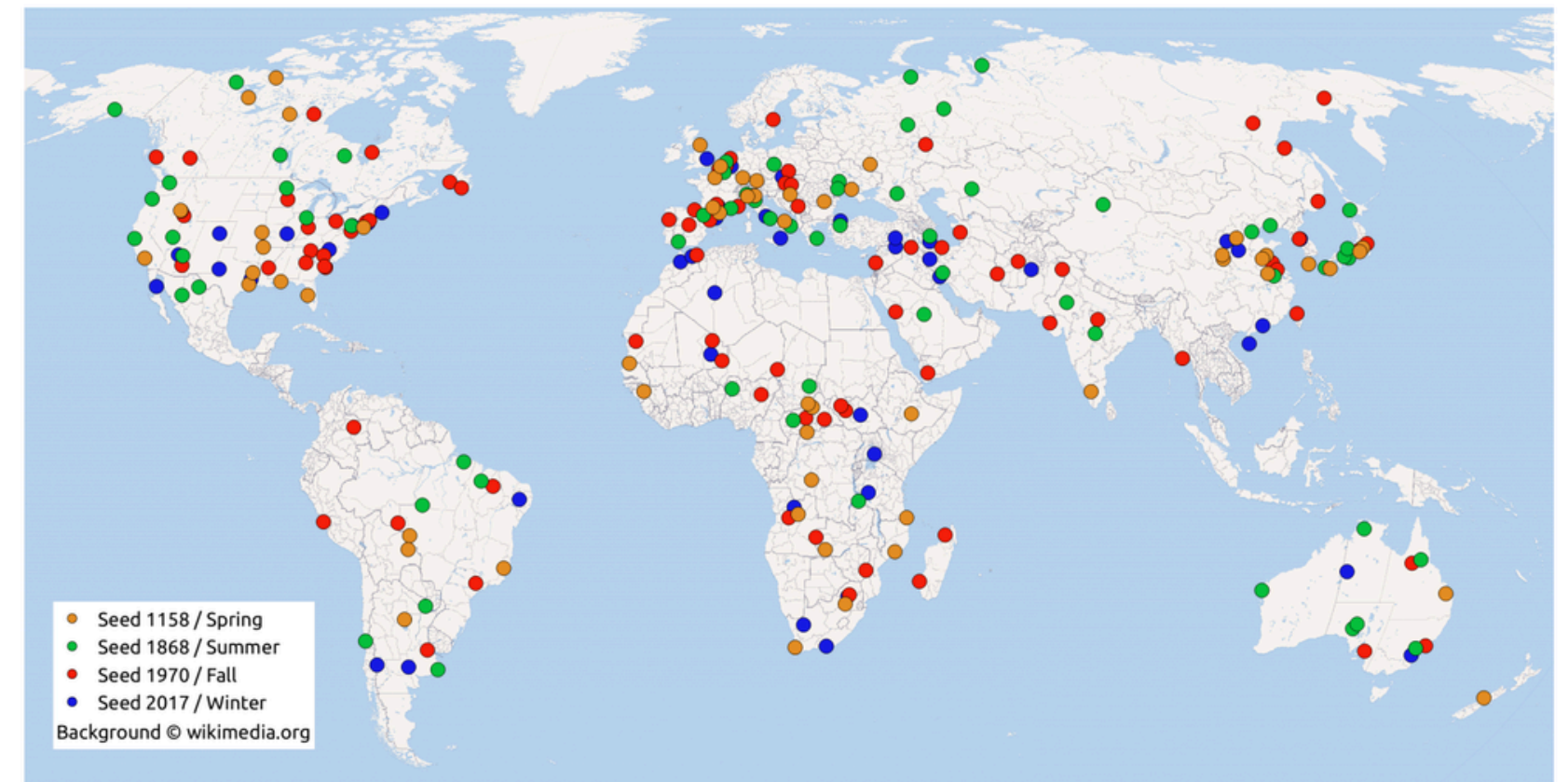
$$\hat{\theta} = \arg \min_{\theta} \mathcal{L}(f_{\theta}(X), Y)$$

Dataset

SEN12-MS

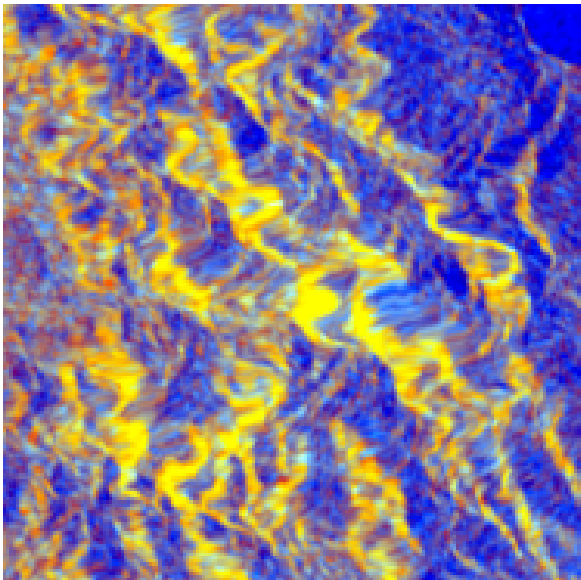
- 180,662 georeferenced image triplets: SAR (vv+vh), cloud free optical (13 bands), and land cover maps
- Each triplet is stored as a 256×256
- Covers all inhabited continents and four meteorological seasons of 2017
- ~ 510 GB in size

[2]

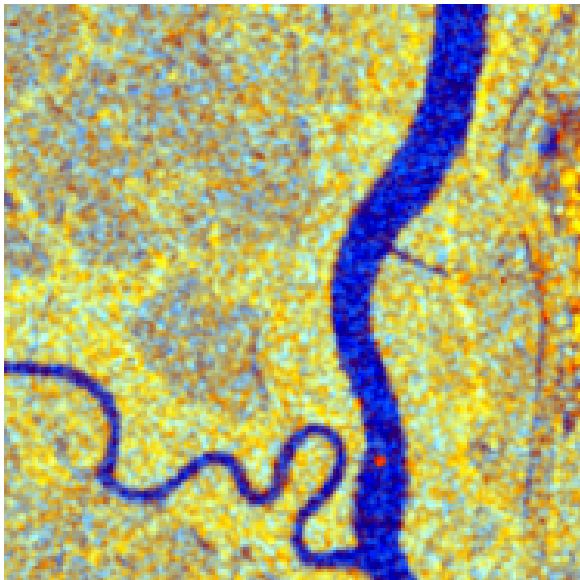


Dataset

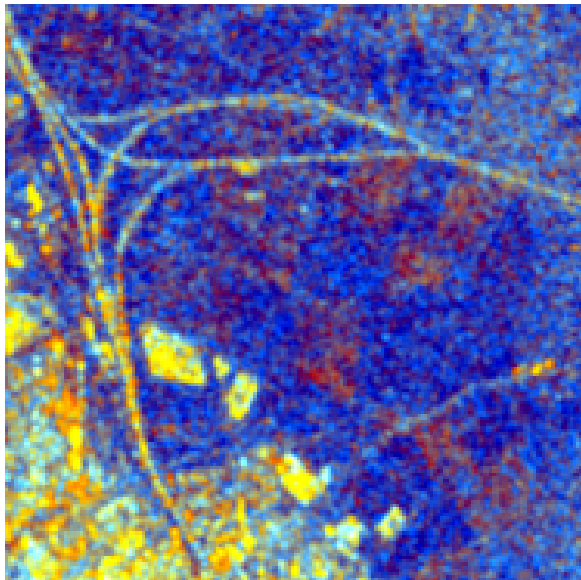
SEN12-MS samples



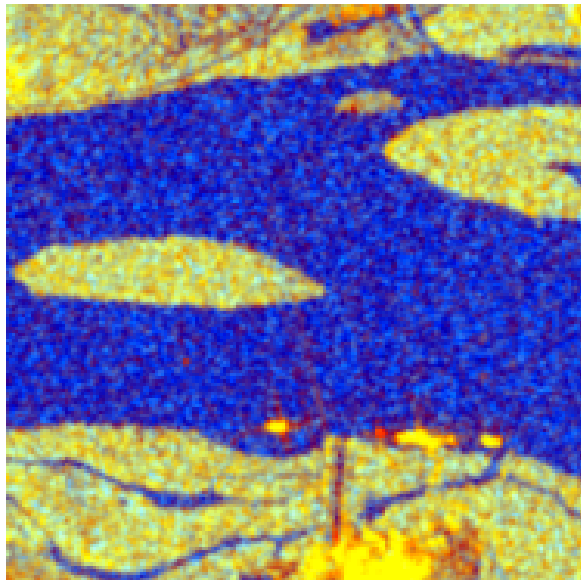
Winter ROI-68-100



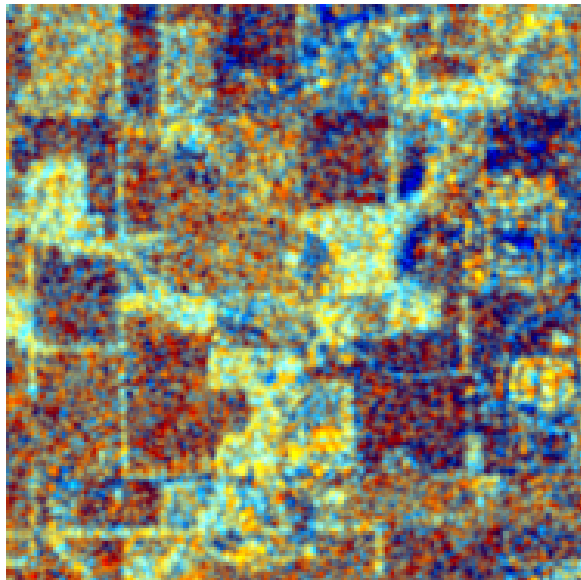
Fall ROI-105-100



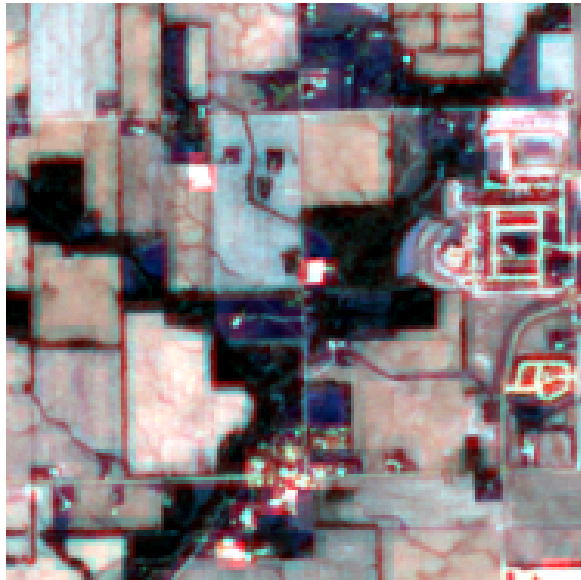
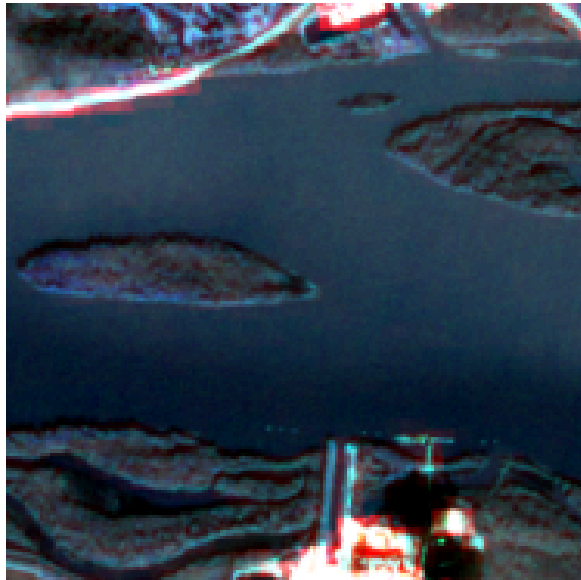
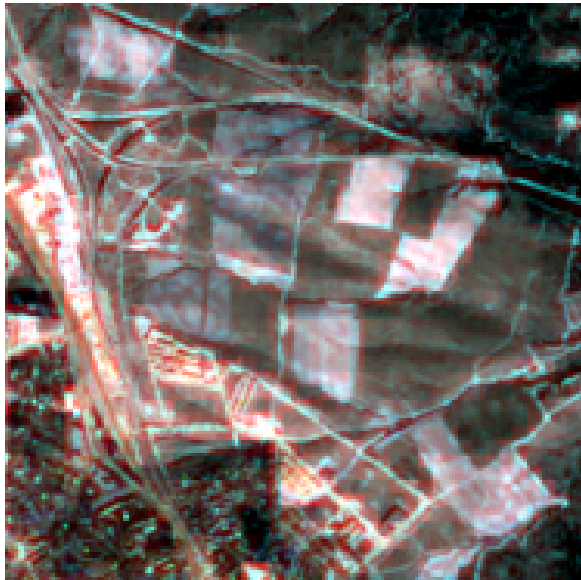
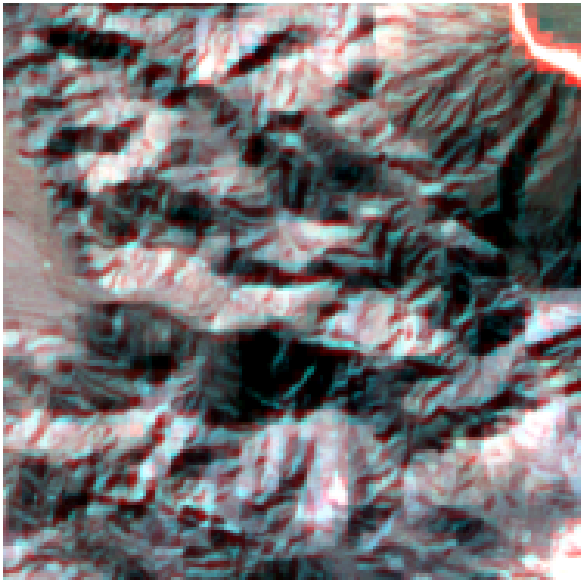
Fall ROI-128-100



Winter ROI-104-101



Fall ROI-145-100



Dataset

SEN12 datasets Family

Aspect	SEN1-2 [50]	SEN12MS [49]	SEN12MS-CR [51]	SEN12MS-CR-TS [52]
Year released	2018	2019	2021	2022
Main purpose	Proof-of-concept SAR-optical translation	Multimodal learning and data fusion	Cloud removal with real cloudy/clear pairs	Multi-temporal cloud removal (sequence models)
Modalities	S1 (VV), S2 (RGB)	S1 (VV,VH), S2 (13 bands), MODIS LULC	S1 (VV,VH), S2 (13 bands; cloudy & cloud-free)	S1 (VV,VH), S2 (13 bands; cloudy & cloud-free time series)
Georeferencing	Not georeferenced	Fully georeferenced	Fully georeferenced	Fully georeferenced
Spatial sampling	Global patch pairs (282k)	180,662 patch triplets across 2017 seasons	169 ROIs; >100k patch triplets	53 ROIs; 30 time steps per ROI
Temporal coverage	Single time-point	Seasonal (2017)	Seasonal with paired cloudy/clear	Year-long time series (2018)
Patch size	256 × 256 px	256 × 256 px	256 × 256 px	256 × 256 px
Notable limitations	RGB only; VV only; no geocoding	MODIS labels are coarse (upsampled)	Mono-temporal pairs (no full time series)	Fewer ROIs; large storage (~2 TB)

Subset Selection & Preprocessing

Training: winter subset (smallest), containing 31,825 paired images

Clipping:

- SAR: VV $\rightarrow$ $[-25\text{dB}, 0]$ VH $\rightarrow$ $[-32.5 \text{ dB}, 0]$
- Optical: $[0, 10000]$

Normalization: $[-1, 1]$ to match the tanh head

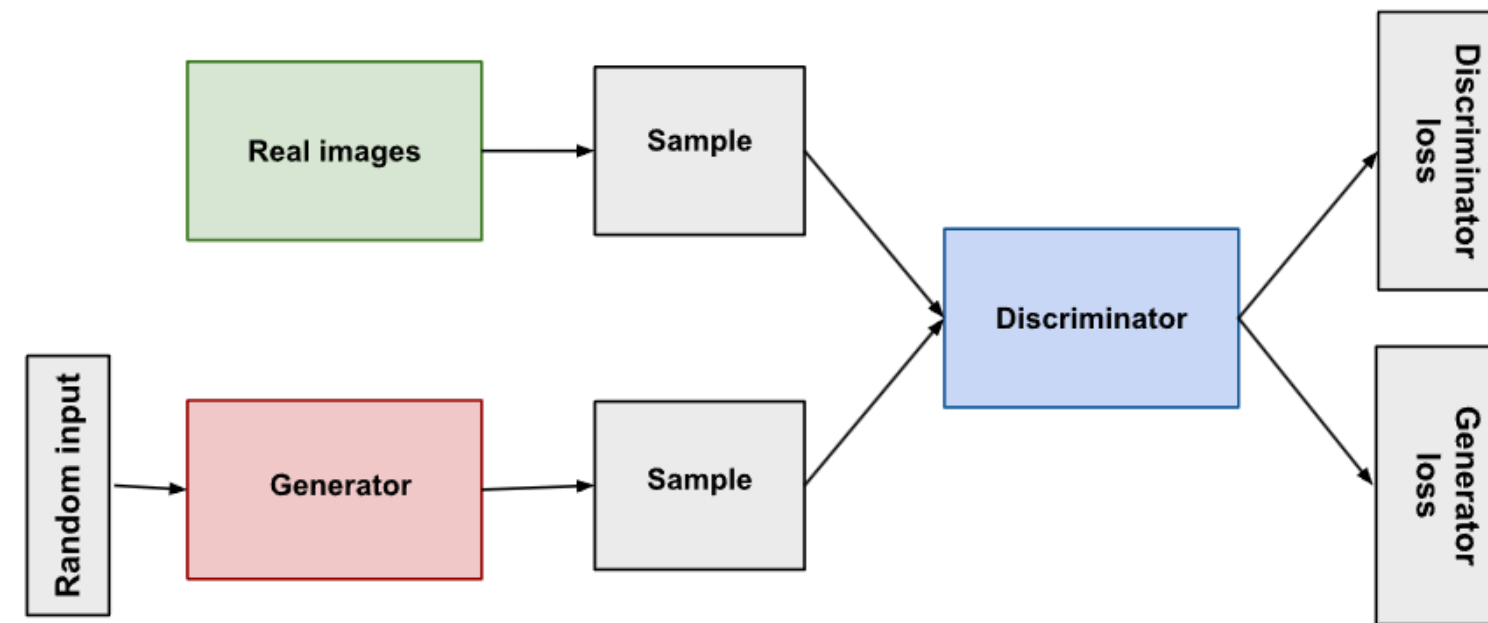
Resolution: 256x256

Augmentation: No data augmentation applied

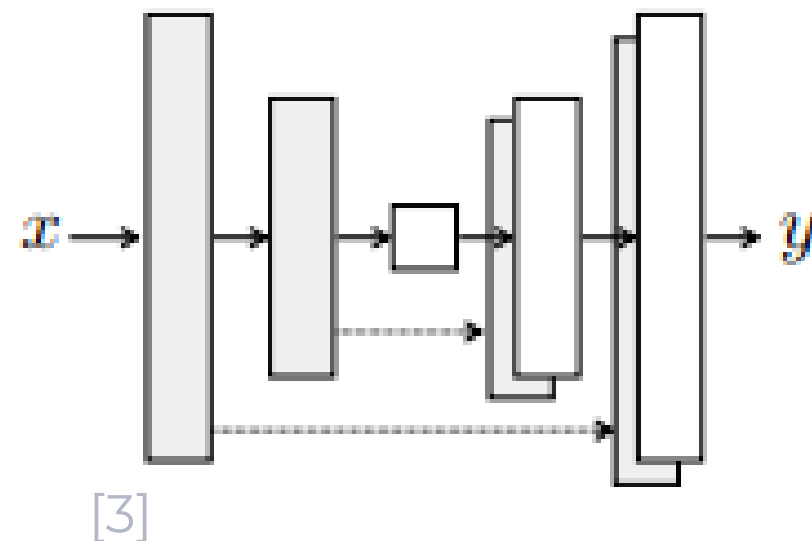
Model

- **Pix2Pix:** cGAN-based model

Source: <https://developers.google.com/>



Generator: U-Net



Generator: PatchGAN

- classifies local $N \times N$ patches of an image as real or fake instead of operating on the entire image

Evaluation Metrics

- Across 16 papers

Metric	References	Frequency
Structural Similarity Index Measurement (SSIM) [57]	[10, 30, 33, 35, 39, 41, 58–63]	12
Peak Signal-to-Noise Ratio (PSNR) [64]	[10, 30, 31, 35, 37, 39, 58–63]	12
Spectral Angle Mapper (SAM) [65]	[8, 31, 35, 41, 58–62, 66]	11
Fréchet Inception Distance (FID) [67]	[30, 33, 39, 61, 62, 66]	6
Root Mean Square Error (RMSE)	[8, 10, 31, 41, 58]	5
Learned Perceptual Image Patch Similarity (LPIPS) [68]	[10, 35, 39, 63, 66]	5
Mean Absolute Error (MAE)	[31, 58]	2
Mean Square Error (MSE)	[37, 60]	2

Evaluation Metrics

SSIM (Structural Similarity Index Measurement):

- compares local patterns of luminance, contrast, and structure between two images
- Values range = $[0, 1]$, the closer to 1, the more similar

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)}$$

PSNR (Peak Signal-to-Noise Ratio):

- quantifies the distortion between a reconstructed image and its reference
- Higher PSNR values indicate lower distortion and better image quality

$$\text{PSNR}(x, y) = 10 \cdot \log_{10} \left(\frac{MAX^2}{\text{MSE}(x, y)} \right), \quad \text{MSE}(x, y) = \frac{1}{N} \sum_{i=1}^N (x_i - y_i)^2,$$

SAM (Spectral Angle Mapper):

- quantifies similarity by measuring the angle between the generated and reference spectral vectors
- smaller values indicating higher spectral similarity and less distortion.

$$\text{SAM}(x, y) = \arccos \left(\frac{\langle x, y \rangle}{\|x\|_2 \cdot \|y\|_2} \right)$$

LPIPS (Learned Perceptual Image Patch Similarity):

- compares feature activations from pretrained convolutional networks (on RGB bands)
- reflects perceptual similarity and visual realism.
- lower LPIPS values indicate that the generated image is closer to the reference in terms of human perceived quality

Evaluation Metrics comparison

Metric	Aspect Evaluated	Advantages	Limitations
PSNR	Pixel-level fidelity via mean squared error ratio	Simple, widely used, interpretable in terms of noise or distortion	Correlates weakly with human perception; sensitive to pixel shifts
SSIM	Structural similarity (luminance, contrast, texture)	Captures perceptual structure better than PSNR; patch-based	Still intensity-based; limited correlation with perceptual realism
SAM	Spectral fidelity across bands	Invariant to illumination; critical for multi-spectral data integrity	Ignores spatial and structural context; only reflects spectral angle
MAE & RMSE	Pixel-wise reconstruction accuracy	Provide direct quantitative estimates of average and large-magnitude errors; easy to interpret	Do not account for perceptual or structural quality; sensitive to scaling and outliers (especially RMSE)
LPIPS	Perceptual similarity using deep feature representations	Aligns well with human judgment; sensitive to high-level semantic differences	Requires pretrained CNN; limited to 3-channel inputs unless adapted

Experimental Setup

- Python 3.11, PyTorch 2.4.0, and CUDA 12.4 on an NVIDIA RTX A5000 GPU with CUDA acceleration

Loss Function

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{GAN}} + \lambda_{\text{L1}} \mathcal{L}_{\text{L1}} + \lambda_{\text{SSIM}} \mathcal{L}_{\text{SSIM}} + \lambda_{\text{LPIPS}} \mathcal{L}_{\text{LPIPS}}.$$

$$\lambda_{\text{L1}} = 100.0, \quad \lambda_{\text{SSIM}} = 50.0, \quad \text{and} \quad \lambda_{\text{LPIPS}} = 50.0$$

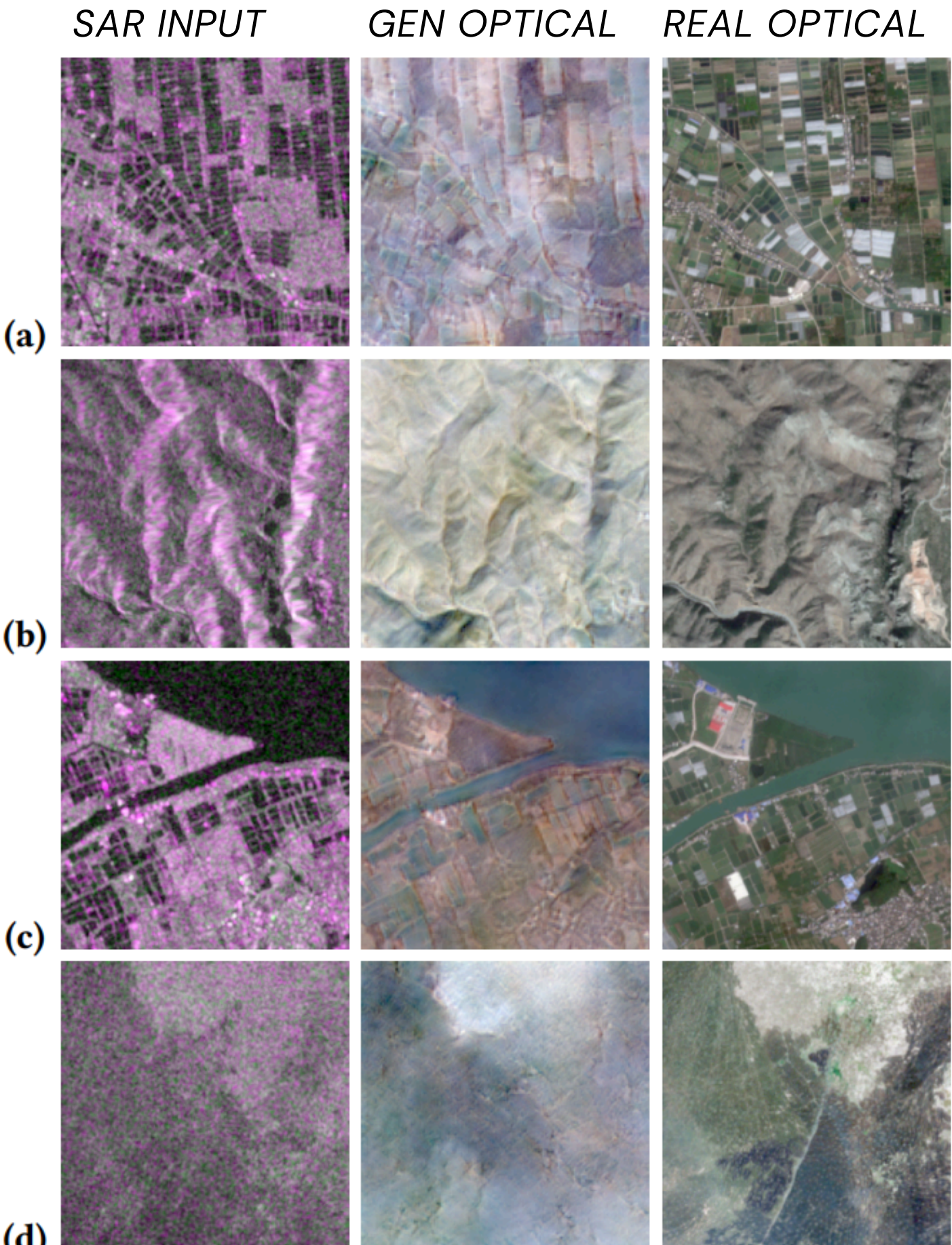
Training Strategy

- 16 batch size
- 150 epochs
- Generator warm up for 20 epochs, then 1:2 update ratio for G and D
- LR: 1e-4 with ReduceLROnPlateau scheduler only for G, patience 10 epochs. D kept constant
- Adam optimizer with $(\beta_1, \beta_2) = (0.5, 0.999)$
- Visual monitoring

RESULTS ON 20% OF THE WINTER SUBSET

- data split: 3,215 train – 981 validation – 981 test

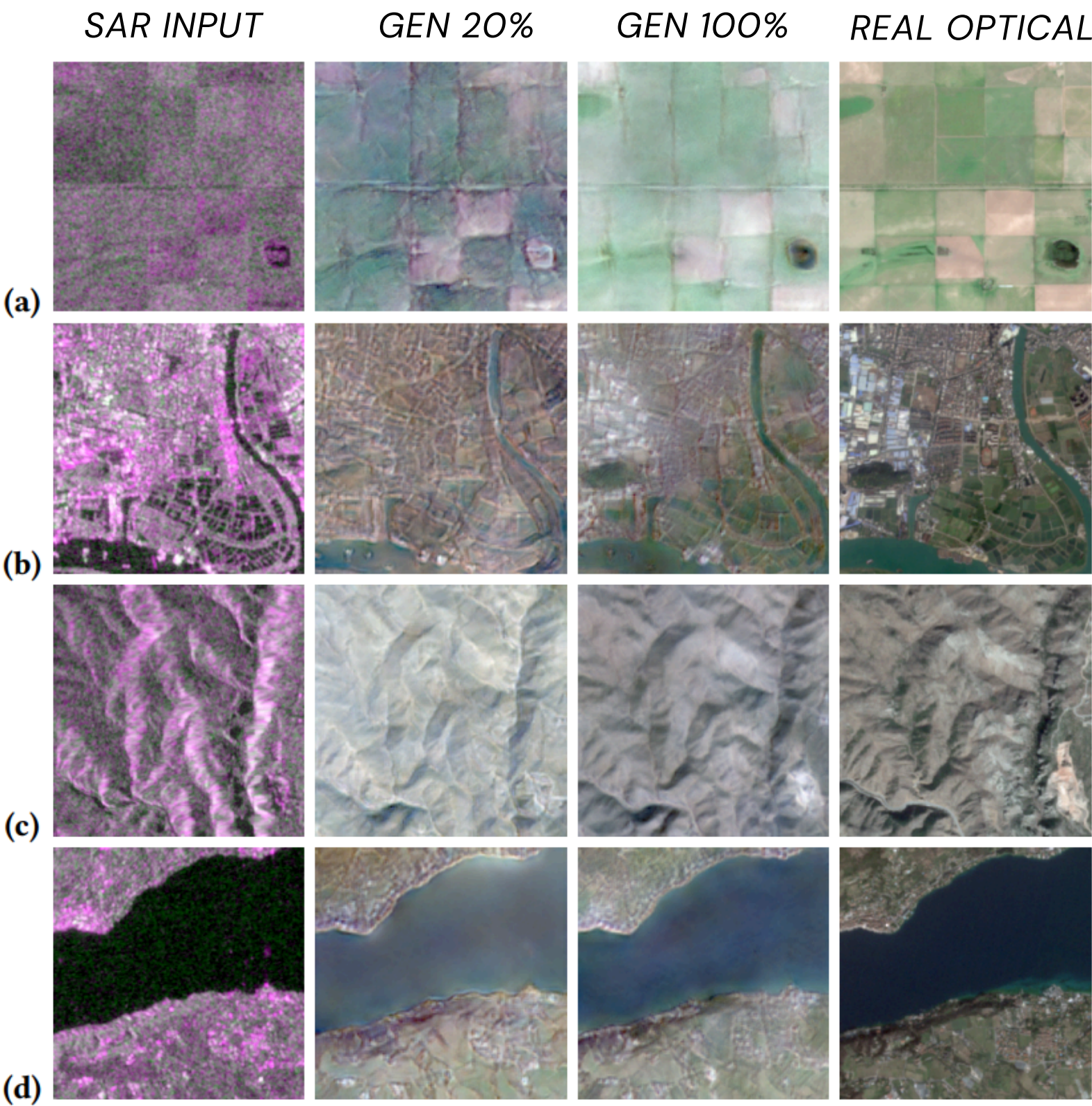
SSIM	PSNR (dB)	LPIPS	SAM (°)	MAE	RMSE
0.859	27.65	0.224	6.71	195.30	381.57



RESULTS ON 100% OF THE WINTER SUBSET

- data split: 31,825 train – 3,180 validation – 3,180 test
- ~25 h for 150 epochs

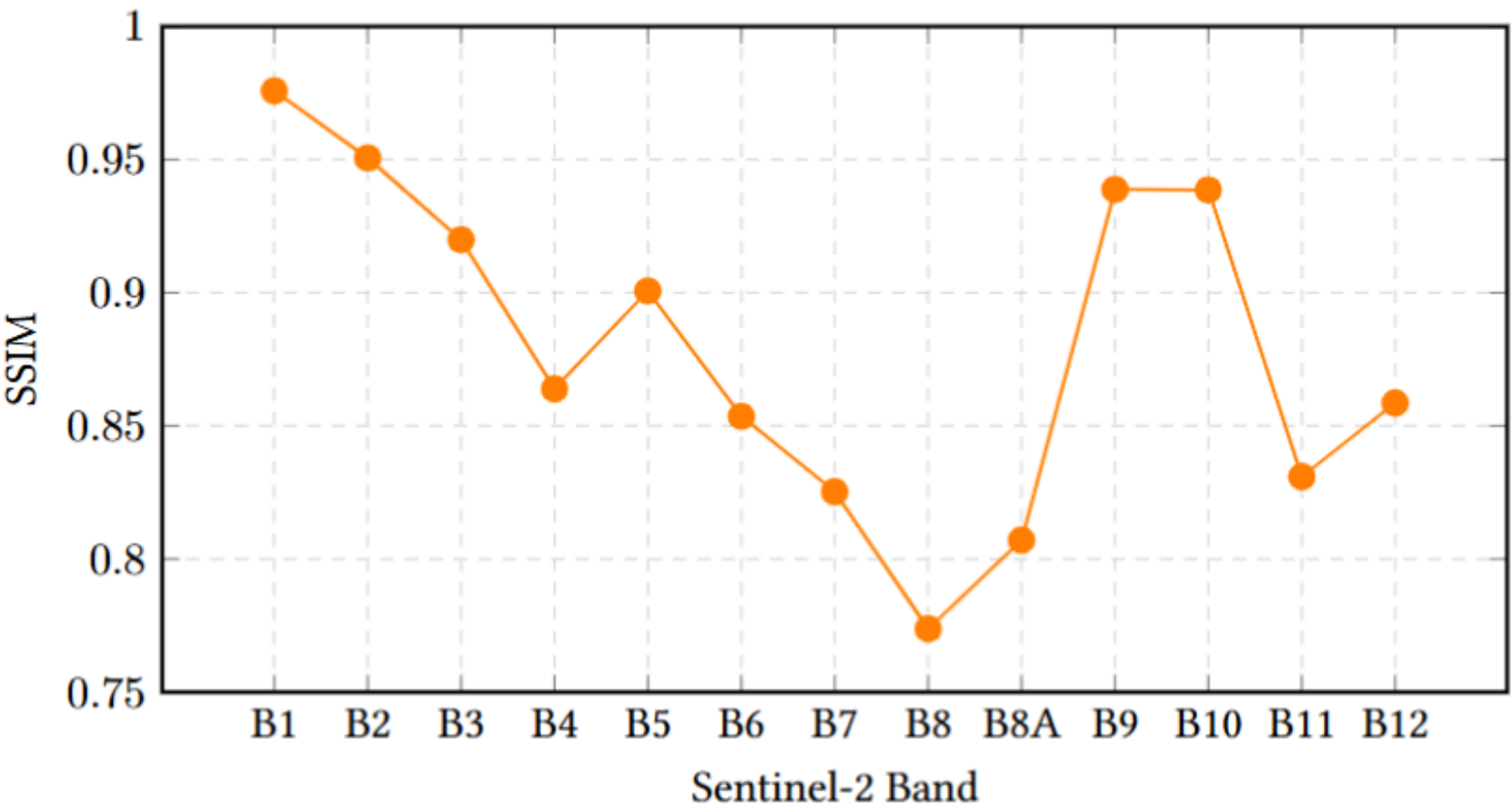
Training Data	SSIM $\uparrow$	PSNR (dB) $\uparrow$	LPIPS $\downarrow$	SAM ($^{\circ}$) $\downarrow$	MAE $\downarrow$	RMSE $\downarrow$
20% of subset	0.859	27.65	0.224	6.71	195.30	381.57
100% of subset	0.888	32.63	0.173	4.41	140.72	233.69



RESULTS ACROSS BANDS

- Model trained on 100% of the winter subset

Band	PSNR (dB) ↑	SSIM ↑	Central Wavelength [nm]	Spectral / Resolution [m]
B1	36.53	0.9758	443	Aerosols / 60
B2	37.49	0.9506	490	Blue / 10
B3	35.67	0.9199	560	Green / 10
B4	32.84	0.8639	665	Red / 10
B5	33.68	0.9007	705	Red Edge / 20
B6	31.62	0.8536	740	Red Edge / 20
B7	30.37	0.8253	783	Red Edge / 20
B8	29.62	0.7738	842	NIR / 10
B8A	29.62	0.8071	865	Red Edge / 20
B9	33.99	0.9388	945	Water Vapour / 60
B10	32.12	0.9386	1375	Cirrus / 60
B11	29.92	0.8309	1610	SWIR / 20
B12	31.48	0.8586	2190	SWIR / 20

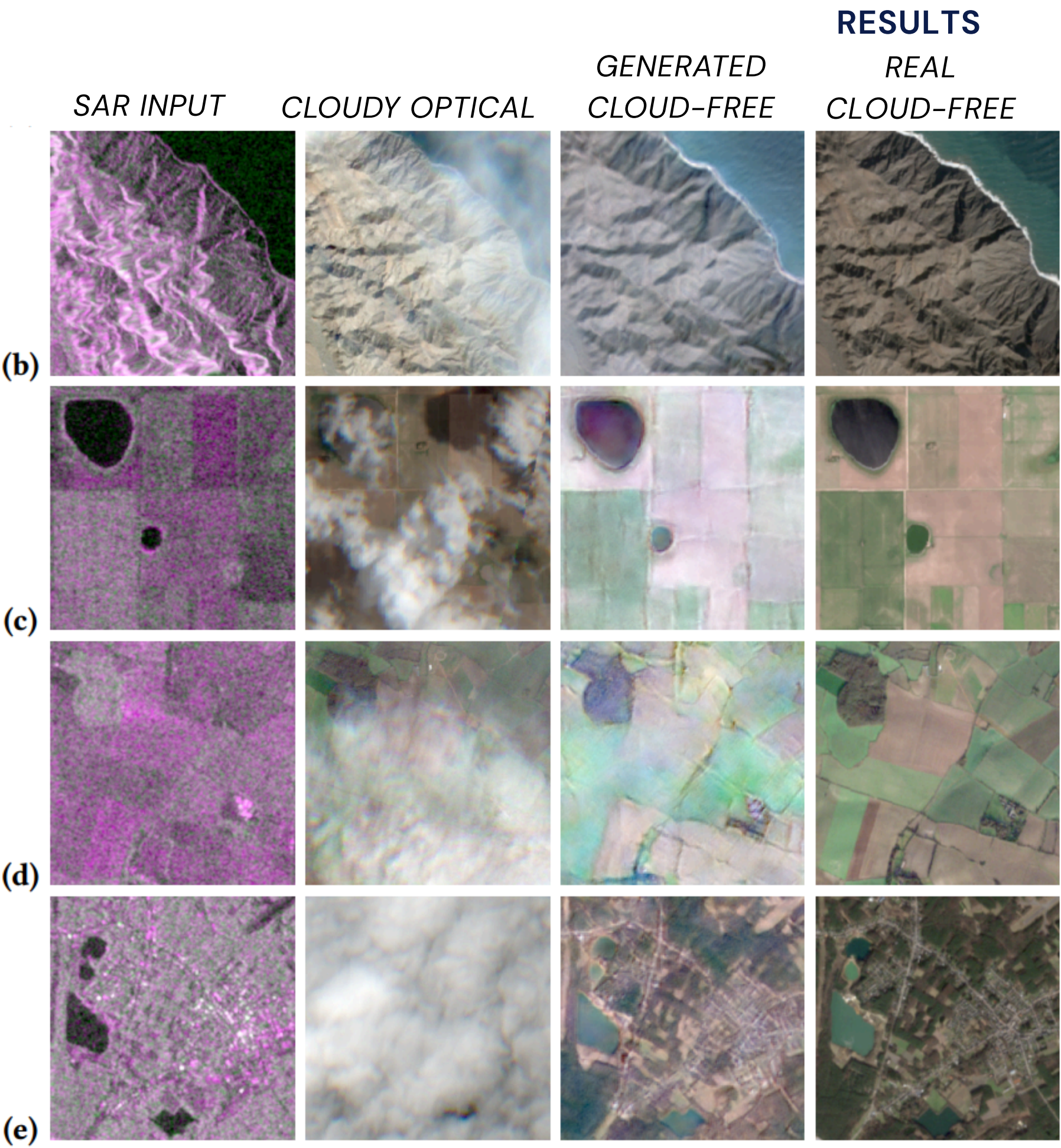


- Model could be biased against B8, since trained only on winter!

RESULTS ON CLOUD REMOVAL

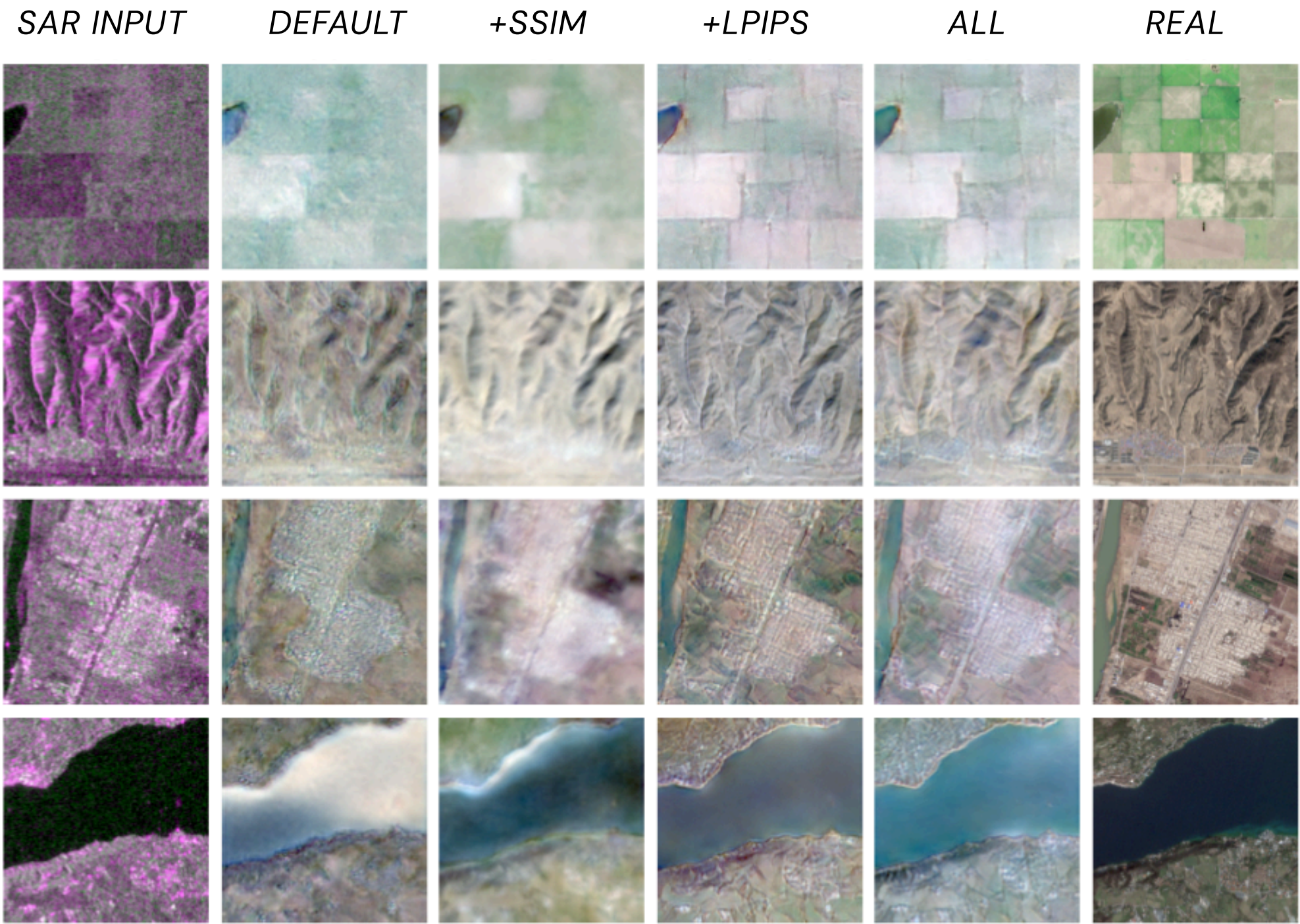
- Model trained on 100% of the winter subset of SEN12-MS
- Test data: 1000 samples from SEN12-MS-CR

Model	SSIM $\uparrow$	PSNR (dB) $\uparrow$	LPIPS $\downarrow$	SAM ($^\circ$) $\downarrow$
DSen2-CR [9]	0.878	–	–	8.07
FAT [58]	0.880	31.85	–	4.19
DiffCR [39] (SOTA)	0.902	–	–	5.82
Pix2Pix (This Thesis)	0.899	33.65	0.152	3.86



EFFECT OF LOSS FUNCTIONS

- 1. $\mathcal{L}_{\text{GAN}} + \mathcal{L}_{\text{L1}}$,
- 2. $\mathcal{L}_{\text{GAN}} + \mathcal{L}_{\text{L1}} + \mathcal{L}_{\text{SSIM}}$,
- 3. $\mathcal{L}_{\text{GAN}} + \mathcal{L}_{\text{L1}} + \mathcal{L}_{\text{LPIPS}}$, and
- 4. the full combination $\mathcal{L}_{\text{GAN}} + \mathcal{L}_{\text{L1}} + \mathcal{L}_{\text{SSIM}} + \mathcal{L}_{\text{LPIPS}}$.



Loss Configuration	SSIM $\uparrow$	PSNR (dB) $\uparrow$	LPIPS $\downarrow$	SAM ($^{\circ}$) $\downarrow$	MAE $\downarrow$	RMSE $\downarrow$
$\mathcal{L}_{\text{GAN}} + \mathcal{L}_{\text{L1}}$	0.820	26.38	0.287	7.88	229	441
$\mathcal{L}_{\text{GAN}} + \mathcal{L}_{\text{L1}} + \mathcal{L}_{\text{SSIM}}$	0.862	27.67	0.399	6.67	198	380
$\mathcal{L}_{\text{GAN}} + \mathcal{L}_{\text{L1}} + \mathcal{L}_{\text{LPIPS}}$	0.842	27.58	0.213	7.05	201	385
$\mathcal{L}_{\text{GAN}} + \mathcal{L}_{\text{L1}} + \mathcal{L}_{\text{SSIM}} + \mathcal{L}_{\text{LPIPS}}$	0.859	27.65	0.224	6.71	195	382

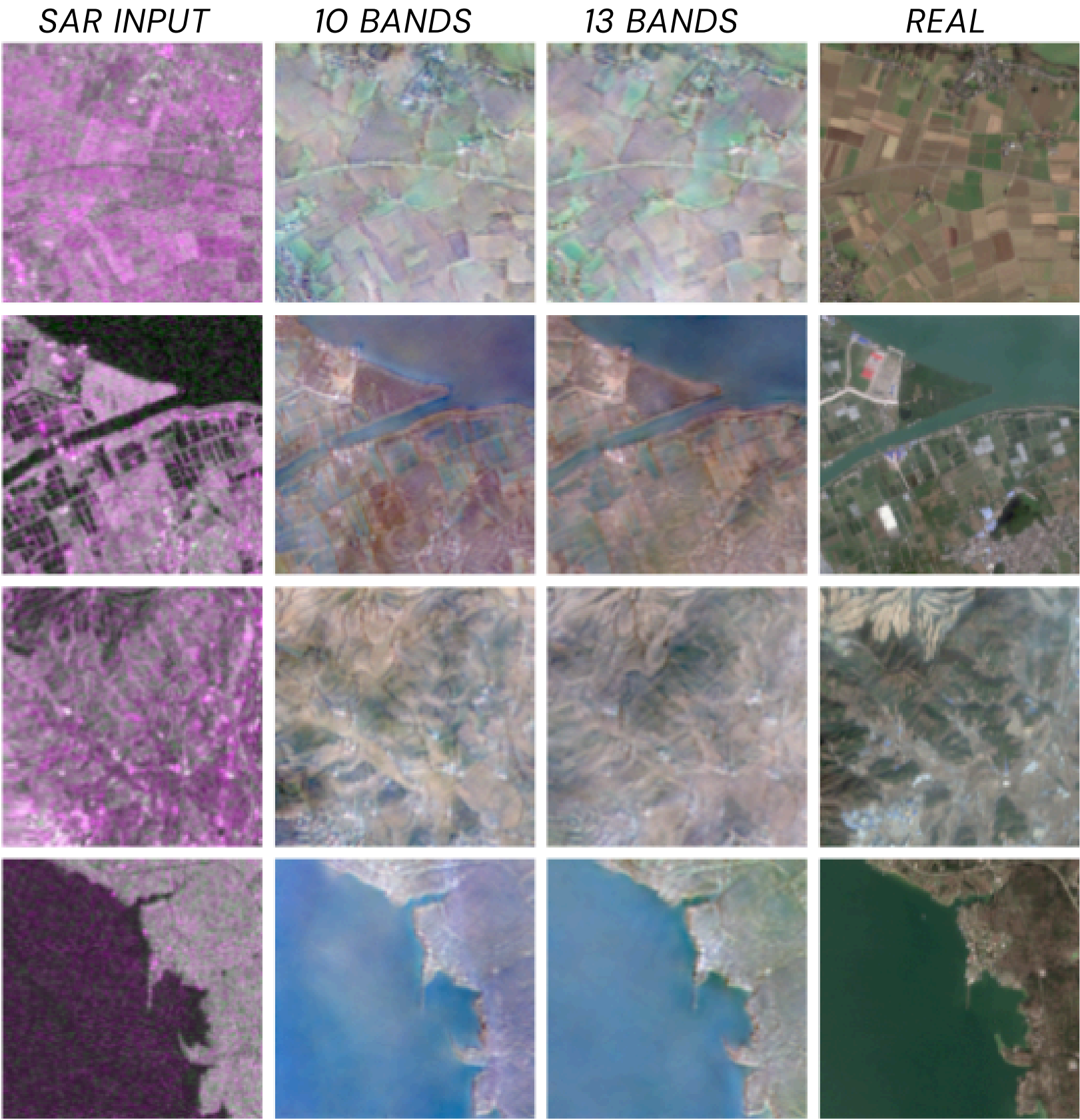
EFFECT OF EXCLUDING 60 M BANDS

- 20% of the winter subset, same training pipeline
- **Hypothesis:** Excluding 60m bands → higher quality ❌

Setting	SSIM ↑	PSNR (dB) ↑	SAM (°) ↓
All 13 bands	0.859390	27.650251	6.7064
Excluding 60m (B1, B9, B10)	0.825861	26.474423	6.3339

- Note: SAM

ABLATION STUDIES



INHERENT MODEL CHARACTERISTICS

GANs are hard to train & Sensitive to hyperparameters.

VANISHING GRADIENTS

- Use LSGAN instead of BCE, since L1 is not optimal for MS generation
- Incorporating SSIM & LPIPS Losses

CHECKERBOARD ARTIFACTS

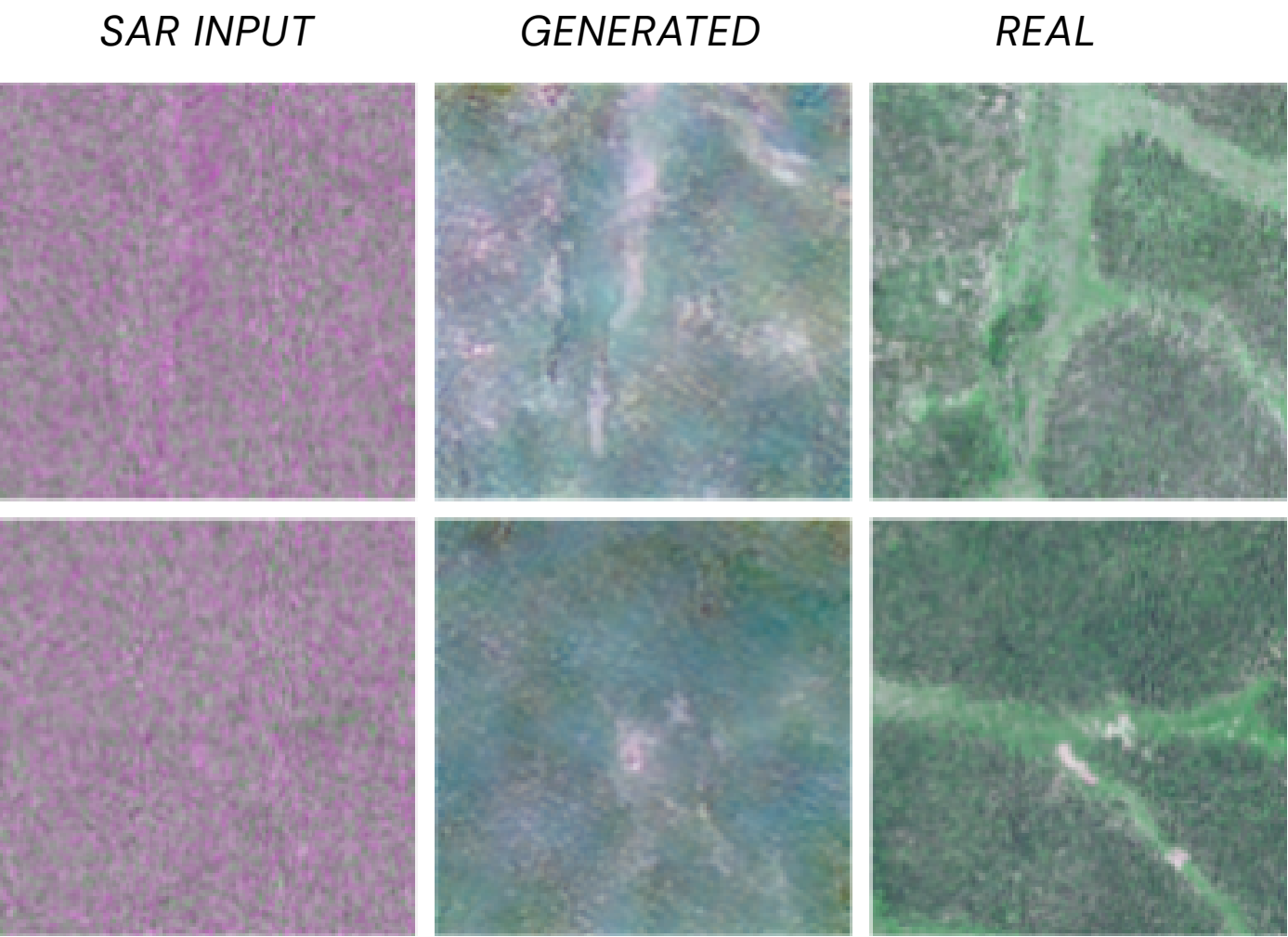
- Transposed convolution layers replaced by nearest-neighbor upsampling followed by standard convolution operations
- deconvolution can easily have “uneven overlap” when the kernel size is not divisible by the stride

```
# Original implementation using ConvTranspose2d
nn.ConvTranspose2d(
    inchannels = ngf * mult,
    outchannels = int(ngf * mult / 2),
    kernelsize = 4,
    stride = 2,
    padding = 1,
    bias = usebias
)
```

→

```
# Replaced with nearest-neighbor upsampling followed by regular
convolution
nn.Upsample(scalefactor = 2, mode = 'bilinear'),
nn.ReflectionPad2d(1),
nn.Conv2d(
    inchannels = ngf * mult,
    outchannels = int(ngf * mult / 2),
    kernelsize = 3,
    stride = 1,
    padding = 0
)
```

Model-Specific Limitations of GAN-Based Translation



Temporal Generalizability Across Different Seasons

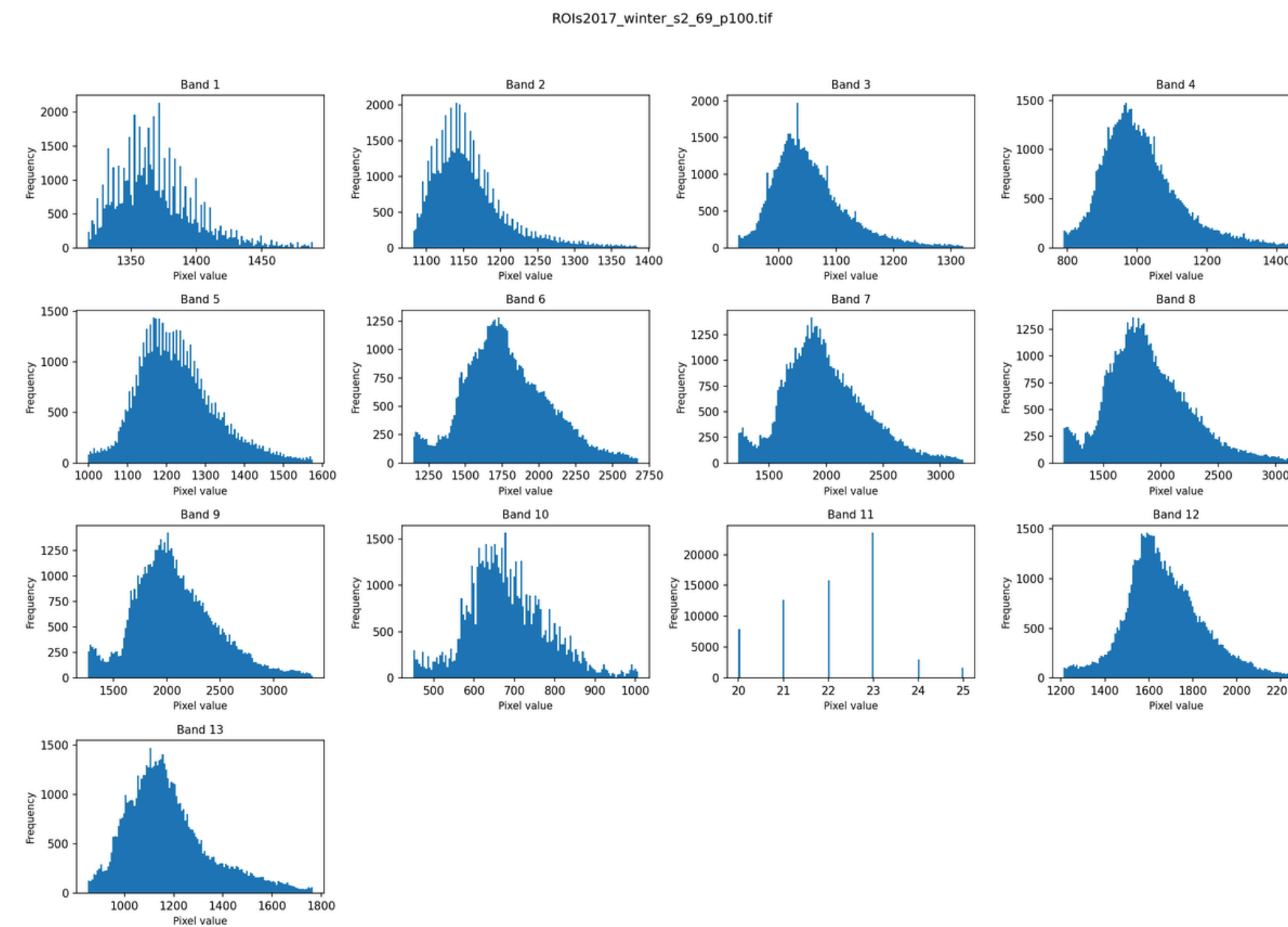
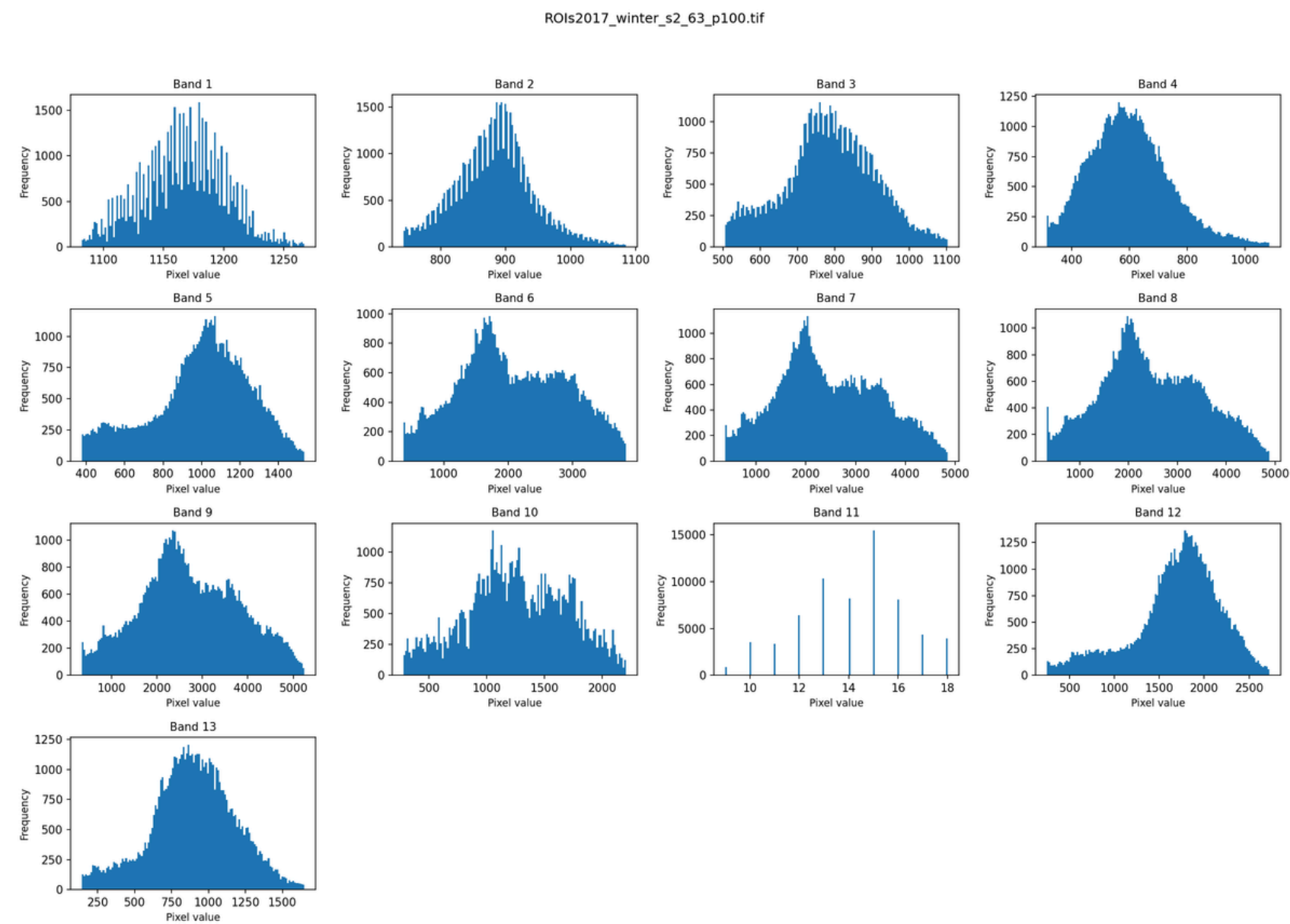
Dataset	SSIM ↑	PSNR (dB) ↑	LPIPS ↓	SAM (°) ↓
Winter	0.888	32.63	0.173	4.41
Spring	0.791	20.00	0.297	12.06
Summer	0.800	22.79	0.268	12.43
Fall	0.788	22.40	0.269	13.67

- Train on the whole dataset
- Train on SEN12-MS-CR

- More research on excluding specific bands

→ More emphasis on Diffusion Models, promising

Aware per-band clipping



- per-band percentiles clipping
- handle 60m bands differently, or drop them

- Validated SAR-to-Optical image translation for 13 bands
- + its use in cloud removal (new & high results)
- Provided across band Evaluation
- Assessed & incorporated SSIM and LPIPS losses, combination resulted in higher performance (new)
- Compared the effect of excluding the 60m bands (new, to the best of my knowledge)
- Illustrated the varying optical data distributions across bands, suggesting aware clipping (new)

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THANK YOU

For your attention

